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Linear LED illumination device with improved color mixing

Abstract

A linear multi-color LED illumination device that produces uniform color throughout the output light beam without the use of excessively large optics or optical losses is disclosed herein. Embodiments for improving color mixing in the linear illumination device include, but are not limited to, a shallow dome encapsulating a plurality of emission LEDs within an emitter module, a unique arrangement of a plurality of such emitter modules in a linear light form factor, and special reflectors designed to improve color mixing between the plurality of emitter modules. In addition to improved color mixing, the illumination device includes a light detector and optical feedback for maintaining precise and uniform color over time and/or with changes in temperature. The light detector is encapsulated within the shallow dome along with the emission LEDs and is positioned to capture the greatest amount of light reflected by the dome from the LED having the shortest emission wavelength.

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Background/Summary

.Iadd.CROSS-REFERENCE TO .Iaddend.RELATED APPLICATIONS

(1) .Iadd.Notice: More than one reissue application has been filed for the reissue of U.S. Pat. No. 9,360,174. The reissue applications are the present application, application Ser. No. 17/590,752, and application Ser. No. 16/001,523 (now U.S. Pat. No. RE48,922). .Iaddend.This application is .Iadd.an application for reissue of U.S. Pat. No. 9,360,174, and is a continuation reissue of application Ser. No. 16/001,523, which is an application for reissue of U.S. Pat. No. 9,360,174, issued Jun. 7, 2016 from application Ser. No. 14/097,339, filed Dec. 5, 2013, which is .Iaddend.related to the following applications: U.S. patent application Ser. No. 14/097,355, now U.S. Pat. No. 9,146,028; U.S. patent application Ser. Nos. 13/970,944; now issued as U.S. Pat. No. 9,237,620; 13/970,964; 13/970,990; 12/803,805; and 12/806,118 now issued as U.S. Pat. No. 8,772,336; each of which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) 1. Field of the Invention

(3) The invention relates to the addition of color mixing optics and optical feedback to produce uniform color throughout the output light beam of a multi-color linear LED illumination device.

(4) 2. Description of Related Art

(5) Multi-color linear LED illumination devices (also referred to herein as lights, luminaires or lamps) have been commercially available for many years. Typical applications for linear LED illumination devices include wall washing in which a chain of lights attempt to uniformly illuminate a large portion of a wall, and cove lighting in which a chain of lights typically illuminates a large portion of a ceiling Multi-color linear LED lights often comprise red, green, and blue LEDs, however, some products use some combination of red, green, blue, white, and amber LEDs.

(6) A multi-color linear LED illumination device typically includes one or more high power LEDs, which are mounted on a substrate and covered by a hemispherical silicone dome in a conventional package. The light output from the LED package is typically lambertian, which means that the LED package emits light in all directions. In most cases, Total Internal Reflection (TIR) secondary optical elements are used to extract the light emitted from a conventional LED package and focus that light into a desired beam. In order to extract the maximum amount of light, the TIR optics must have a specific shape relative to the dome of the LED package. Other dimensions of the TIR optics determine the shape of the emitted light beam.

(7) Some multi-color linear LED light products comprise individually packaged LEDs and individual TIR optics for each LED. In order for the light emitted from the different colored LED emitters to mix properly, the light beams from each individual color LED must overlap. However, because the LEDs are spaced centimeters apart, the beams will overlap and the colors will mix only in the far field, at some distance away from the linear light. At a very close range to the linear light, the beams will be separate and the different colors are clearly visible. Although such a product may exhibit good color mixing in the far field, it does not exhibit good color mixing in the near field.

(8) Other multi-color linear LED light products use red, green, and blue LEDs packaged together with a single TIR optic attached to each RGB LED package. These RGB LED packages typically comprise an array of three or four LEDs, which are placed as close together as possible on a substrate and the entire array is covered by one hemispherical dome. In products that use one TIR optical element for each multi-color LED package, there is not necessarily a need for the beams from the different TIR optical elements to overlap for the colors to mix. Therefore, such products

tend to have better near field color mixing than products that use individually packaged LEDs.
(9) However, depending on the size of the primary and secondary optics, the far field color mixing may actually be worse in products that package multiple colors of LEDs together. Since the different colored LEDs are in physically different locations within the hemispherical silicone dome, the light radiated from the dome, and therefore, from the TIR optical element will not be perfectly mixed. Although larger domes and larger TIR optical elements may be used to provide better color mixing, there are practical limits to the size of these components, and consequently, to the near and far field color mixing provided by such an approach.
(10) An alternative optical system, although not commonly used, for color mixing and beam shaping in multi-color LED linear lights uses reflectors. In some cases, the light from a plurality of multi-colored LED emitter packages are mixed by a diffusion element and shaped by a concave reflector that redirects the light beams down a wall. The diffusion element could be combined with an exit lens or could be a shell diffuser placed over the multi-color emitter packages, for instance. Alternatively, the system could use a shell diffuser and a diffused exit lens. Although such systems can achieve very good color mixing in both the near and the far field, there is a tradeoff between color mixing and optical efficiency. As the amount of diffusion increases, the color mixing improves, but the optical efficiency decreases as the diffuser absorbs and scatters more light.
(11) As LEDs age, the light output at a given drive current changes. Over thousands of hours, the light output from any individual LED may decrease by approximately 10-25% or more. The amount of degradation varies with drive current, temperature, color, and random defect density. As such, the different colored LEDs in a multi-color LED light will age differently, which changes the color of the light produced by the illumination device over time. A high quality multi-color LED light that can maintain precise color points over time should have the means to measure the light output from each color component, and adjust the drive current to compensate for changes. Further, a multi-color linear light should have the means to measure the light produced by each set of colored LEDs independent from other sets to prevent part of the linear light from producing a different color than other parts.
(12) Multi-color LED linear lights with TIR optics on each individual LED cannot achieve good color mixing in the near field. Multi-color LED linear lights that combine a multi-color LED package with a TIR optical element require a large TIR optical element to achieve good color mixing in the near and far fields. Multi-color LED linear lights that use conventional diffusers and reflectors to achieve good color mixing in both the near and the far field suffer optical losses. As such, there is a need for an improved optical system for multi-color LED linear lights that provides good color mixing in the near and far fields, is not excessively large and expensive, and has good optical efficiency. Further, there is a need for an optical feedback system to maintain precise color in such linear lights. The invention described herein provides a solution.

SUMMARY OF THE INVENTION

(13) A linear multi-color LED illumination device that produces a light beam with uniform color throughout the output beam without the use of excessively large optics or optical losses is disclosed herein. In addition to improved color mixing, the illumination device includes a light detector and optical feedback for maintaining precise and uniform color over time and/or with changes in temperature. The illumination device described herein may also be referred to as a light, luminaire or lamp.
(14) Various embodiments are disclosed herein for improving color mixing in a linear multi-color LED illumination device. These embodiments include, but are not limited to, a uniquely configured dome encapsulating a plurality of emission LEDs and a light detector within an emitter module, a unique arrangement of the light detector relative to the emission LEDs within the dome, a unique arrangement of a plurality of such emitter modules in a linear light form factor, and reflectors that are specially designed to improve color mixing between the plurality of emitter modules. The embodiments disclosed herein may be utilized together or separately, and a variety of features and

variations can be implemented, as desired, to achieve optimum color mixing results. In addition, related systems and methods can be utilized with the embodiments disclosed herein to provide additional advantages or features. Although the various embodiments disclosed herein are described as being implemented in a linear light form factor, certain features of the disclosed embodiments may be utilized in illumination devices having other form factors to improve the color mixing in those devices.

(15) According to one embodiment, an illumination device is disclosed herein as including a plurality of LED emitter modules, which are spaced apart from each other and arranged in a line. Each emitter module may include a plurality of emission LEDs whose output beams combine to provide a wide color gamut and a wide range of precise white color temperatures along the black body curve. For example, each emitter module may include four different colors of emission LEDs, such as red, green, blue, and white LEDs. In such an example, the red, green, and blue emission LEDs may provide saturated colors, while a combination of light from the RGB LEDs and a phosphor converted white LED provide a range of whites and pastel colors. However, the emitter modules described herein are not limited to any particular number and/or color of emission LEDs, and may generally include a plurality of emission LEDs, which include at least two different colors of LEDs. The plurality of LEDs may be arranged in a two-dimensional array (e.g., a square array), mounted on a substrate (e.g., a ceramic substrate), and encapsulated within a dome.

(16) In some embodiments, the linear illumination device may comprise six emitter modules per foot, and each emitter module may be rotated approximately 120 degrees relative to the next adjacent emitter module. The rotation of subsequent emitters in the line improves color mixing between adjacent emitter modules to some degree. Although such an arrangement has been shown to provide sufficient lumen output, efficacy, and color mixing, one skilled in the art would understand how the inventive concepts described herein can be applied to other combinations of LED numbers/colors per emitter module, alternative numbers of LED emitter modules per foot, and other angular rotations between emitter modules without departing from the scope of the invention.

(17) In general, an illumination device in accordance with the present invention may include at least a first emitter module, a second emitter module, and a third emitter module arranged in a line, wherein the second emitter module is spaced equally distant between the first and third emitter modules. To improve color mixing, the second emitter module may be rotated X degrees relative to the first emitter module, and the third emitter module may be rotated 2X degrees relative to the first emitter module. X may be substantially any rotational angle equal to 360 degrees divided by an integer N, where N is greater than or equal to 3.

(18) In some embodiments, color mixing may be further improved by covering each emitter module with an optically transmissive dome, whose shallow or flattened shape allows a significant amount of light emitted by the LED array to escape out of the side of the emitter module. For example, a shallow dome may be formed with a radius in a plane of the LED array that is about 20-30% larger than the radius of the curvature of the shallow dome. Such a shape may enable approximately 40% of the light emitted by the LED array to exit the shallow dome at small angles (e.g., approximately 0 to 30 degrees) relative to the plane of the LED array.

(19) In some embodiments, color mixing may be further improved by the inclusion of a specially designed reflector, which is suspended above the plurality of emitter modules. The reflector comprises a plurality of louvers, each of which may be centered upon and suspended a spaced distance above a different one of the emitter modules. These louvers comprise a substantially circular shape with sloping sidewalls, which are angled so that a top diameter of the louver is substantially larger than a bottom diameter of the louver. The louvers are configured to focus a majority of the light emitted by the emitter modules into an output beam by configuring the bottom diameter of the louvers to be substantially larger than the diameter of the emitter modules. In some cases, the sloping sidewalls of the louvers may include a plurality of planar facets, which

randomize the direction of light rays reflected from the planar facets.
(20) By suspending the louvers a spaced distance above the emitter modules, the louvers allow the portion of the light that emanates sideways from adjacent emitter modules to mix underneath the louvers before that light is redirected out of the illumination device through an exit lens. In some embodiments, the louvers may be suspended approximately 5 mm to approximately 10 mm above the emitter modules. Other distances may be appropriate depending on the particular design of the emitter modules and the louvers.
(21) In some embodiments, an exit lens may be provided with a combination of differently textured surfaces and/or patterns on opposing sides of the lens to further promote color mixing. For example, an internal surface of the exit lens may comprise a flat roughened surface that diffuses the light passing through the exit lens. An external surface of the exit lens may comprise an array of micro-lenses, or lenslets, to further scatter the light rays and shape the output beam.
(22) In some embodiments, each emitter module may also comprise a detector, which is configured to detect light emitted by the emission LEDs. The detector is mounted onto the substrate and encapsulated within the shallow dome, along with the emission LEDs, and may be an orange, red or yellow LED, in one embodiment. Regardless of color, the detector LED is preferably placed so as to receive the greatest amount of reflected light from the emission LED having the shortest wavelength. For example, the emission LEDs may include red, green, blue and white LEDs arranged in a square array, in one embodiment. In this embodiment, the detector LED is least sensitive to the shortest wavelength emitter LED, i.e., the blue LED. For this reason, the detector LED is positioned on the side of the array that is furthest from the blue LED, so as to receive the greatest amount of light reflected off the dome from the blue LED. In some cases, the dome may have a diffuse or textured surface, which increases the amount of light that is reflected off the surface of the dome back towards the detector LED.
(23) In addition to the emitter modules, the illumination device described herein includes a plurality of driver circuits coupled to the plurality of LEDs for supplying drive currents thereto. During a compensation period, the plurality of driver circuits are configured to supply drive currents to the plurality of emission LEDs, one LED at a time, so that the detector LED can detect the light emitted by each individual LED. A receiver is coupled to the detector LED for monitoring the light emitted by each individual LED and detected by the detector LED during the compensation period. In some embodiments, the receiver may comprise a trans-impedance amplifier that detects the amount of light produced by each individual LED. Control logic is coupled to the receiver and the driver circuits for controlling the drive currents produced by the driver circuits based on the amount of light detected from each LED. In some embodiments, the control logic may use optical and/or temperature measurements obtained from the emission LEDs to adjust the color and/or intensity of the light produced by the illumination device over time and/or with changes in temperature.
(24) Various other patents and patent applications assigned to the assignee, including U.S. Publication No. 2010/0327764, describe means for periodically turning all but one emission LED off during the compensation period, so that the light produced by each emission LED can be individually measured. Other patent applications assigned to the assignee, including U.S. patent application Ser. Nos. 13/970,944; 13/970,964; and 13/970,990 describe means for measuring a temperature of the LEDs and adjusting the intensity of light emitted by the LEDs to compensate for changes in temperature. These commonly assigned patents and patent applications are incorporated by reference in their entirety. The invention described herein utilizes the assignee's earlier work and improves upon the optical measurements by placing the detector LED within the dome, and away from the shortest wavelength LED, to ensure the light for all emission LEDs is properly detected.
(25) Any detector in a multi-color light source with optical feedback should be placed to minimize interference from external light sources. This invention places the detectors within the silicone dome to prevent interference from external sources and other emitter modules within the linear

light. The detectors are preferably red, orange or yellow LEDs, but could comprise silicon diodes or any other type of light detector. However, red, orange or yellow detector LEDs are preferable over silicon diodes, since silicon diodes are sensitive to infrared as well as visible light, while LEDs are sensitive to only visible light.
(26) In some embodiments, the illumination device may further include an emitter housing, a power supply housing coupled to the emitter housing and at least one mounting bracket for mounting the illumination device to a surface (e.g., a wall or ceiling) The emitter modules, the reflector and the driver circuits described above reside within the emitter housing. The exit lens is mounted above the reflector and attached to sidewalls of the emitter housing. In some embodiments, the power supply housing may be coupled to a bottom surface of the emitter housing and comprises an orifice through which a power cable may be routed and connected to a power converter housed within the power supply housing. In some embodiments, a special hinge mechanism may be coupled between the emitter housing and the at least one mounting bracket. As described in the commonly assigned co-pending U.S. application Ser. No. 14/097,335, the hinge mechanism allows the emitter housing to rotate approximately 180 degrees relative to the mounting bracket around a rotational axis of the hinge mechanism. The co-pending application is hereby incorporated in its entirety.

Description

DESCRIPTION OF THE DRAWINGS
(1) Other objects and advantages of the invention will become apparent upon reading the following detailed description and upon reference to the accompanying drawings.
(2) FIG. **1** is a picture of an exemplary full color gamut linear LED light.
(3) FIG. **2** is an exemplary illustration of the rotating hinges shown in FIG. **1**.
(4) FIG. **3** provides additional illustration of the rotating hinge components.
(5) FIG. **4** is a picture of exemplary components that may be included within the full color gamut linear LED light of FIG. **1**.
(6) FIG. **5** is an exemplary block diagram of circuitry that may be included on the driver board and the emitter board of the exemplary full color gamut linear LED light of FIG. **1**.
(7) FIG. **6** is an exemplary block diagram of the interface circuitry and emitter module of FIG. **5**.
(8) FIG. **7** is an illustration of an exemplary color gamut that may be produced by the linear LED light on a CIE1931 color chart.
(9) FIG. **8** is a photograph of an exemplary LED emitter module comprising a plurality of emission LEDs and a detector LED mounted on a substrate and encapsulated in a shallow dome.
(10) FIG. **9** is a side view drawing of the LED emitter module of FIG. **8**.
(11) FIG. **10**A is a drawing of an exemplary LED emitter module depicting a desirable placement of the emission LEDs and the detector LED within the dome, according to one embodiment.
(12) FIG. **10**B is a drawing of an exemplary LED emitter module depicting another desirable placement of the emission LEDs and the detector LED within the dome, according to another embodiment.
(13) FIG. **11** is a photograph of an exemplary emitter board comprising a plurality of LED emitter modules, wherein sets of the modules are rotated relative to each other to promote color mixing.
(14) FIG. **12** is a photograph of an exemplary emitter board, emitter housing and reflector for a full color gamut linear LED light with a 120 degree beam angle.
(15) FIG. **13** is a photograph of an exemplary emitter board, emitter housing and a reflector for a full color gamut linear LED light with a 60 degree beam angle.
(16) FIG. **14** is an exemplary ray diagram illustrating how the shallow dome of the emitter modules and the reflector of FIG. **13** enable light rays from adjacent emitter modules to mix together to

promote color mixing.
(17) FIG. **15** is an exemplary drawing providing a close up view of one of the emitter modules and floating louvers shown in FIG. **14**.
(18) FIG. **16** is an exemplary drawing of an exit lens comprising a plurality of lenslets formed on an external surface of the lens, according to one embodiment.
(19) FIG. **17** is an exemplary ray diagram illustrating the effect that the exit lens shown in FIG. **16** has on the output beam when the plurality of lenslets formed on the external surface is combined with a textured internal surface.
(20) While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.
DETAILED DESCRIPTION OF THE INVENTION
(21) Turning now to the drawings, FIG. **1** is a picture of a linear LED lamp **10**, according to one embodiment of the invention. As described in more detail below, linear LED lamp **10** produces light over a wide color gamut, thoroughly mixes the color components within the output beam, and uses an optical feedback system to maintain precise color over LED lifetime, and in some cases, with changes in temperature. The linear LED lamp **10** shown in FIG. **1** is powered by the AC mains, but may be powered by alternative power sources without departing from the scope of the invention. The light beam produced by LED lamp **10** can be symmetric or asymmetric, and can have a variety of beam angles including, but not limited to, 120×120, 60×60, and 60×30. If an asymmetric beam is desired, the asymmetric beam typically has a wider beam angle across the length of the lamp.
(22) In general, LED lamp **10** comprises emitter housing **11**, power supply housing **12**, and rotating hinges **13**. As shown more clearly in FIG. **4**, and discussed below, emitter housing **11** comprises a plurality of LED driver circuits, a plurality of LED emitter modules and a reflector, which is mounted a spaced distance above the emitter modules for focusing the light emitted by the emitter modules. The power supply housing **12** comprises an AC/DC converter powered by the AC mains, in one embodiment. Rotating hinges **13** allow both emitter housing **11** and power supply housing **12** to rotate 180 degrees relative to a pair of mounting brackets **14**, which provides installation flexibility. Although a pair of mounting brackets **14** are shown in FIG. **1**, alternative embodiments of the LED lamp may include a greater or lesser number of brackets, as desired.
(23) In linear lighting fixtures, such as LED lamp **10**, one major design requirement is to have the power cable enter and exit through the axis of rotation. This requirement allows adjacent lighting fixtures to be independently adjusted, while maintaining a constant distance between connection points of adjacent lighting fixtures. However, this requirement complicates the design of the rotational hinges used in linear lighting, as it prevents the hinges from both rotating and passing power through the same central axis. LED lamp **10** solves this problem by moving the rotational components of the hinge off-axis, and joining the rotational components to the central axis with a swing arm to a rack and pinion gear assembly. An embodiment of such a solution is shown in FIGS. **2**-**3** and described below.
(24) As shown in FIG. **2**, each rotating hinge **13** may include a swing arm **15**, an end cap **17** and a hinge element **16**. The end cap **17** may be configured with a flat upper surface for attachment to the emitter housing **11** and a semi-circular inner surface comprising a plurality of teeth. One end of the swing arm **15** is securely mounted onto the mounting bracket **14** of the linear LED lamp **10**. In some embodiments, the swing arm **15** can be secured to the mounting bracket **14** by way of screws **19**, as shown in FIG. **3**. However, alternative means of attachment may be used in other embodiments of the invention. An opposite end of the swing arm **15** is coupled near the flat upper

surface of the end cap **17** and is centered about the rotational axis of the hinge mechanism. The opposite end of the swing arm comprises a cable exit gland **18**, which is aligned with the orifice of the power supply housing for routing the power cable into the power supply housing at the rotational axis of the hinge mechanism.

(25) As shown in FIGS. **2** and **3**, swing arm **15** houses a hinge element **16** that provides an amount of resistance needed to secure the lamp **10** in substantially any rotational position within a 180 degree range of motion. The hinge element **16** extends outward from within the swing arm **15** and generally comprises a position holding gear, which is configured to interface with the toothed end cap **17** of the linear LED lamp **10**. In some embodiments, the hinge element **16** may further comprise a constant torque element that provides a substantially consistent amount of torque to the position holding gear, regardless of whether the position holding gear is stationary or in motion. In other embodiments, the constant torque element may be replaced with a high static energy/low kinetic energy rotational element to enable easier rotational adjustment, while still providing the necessary resistance to hold the lamp **10** in the desired rotational position.

(26) The rotating hinge **13** shown in FIGS. **2-3** enables electrical wiring (e.g., a power cable) to be routed through the rotational axis of the rotating hinge **13** and to enter/exit the hinge at the cable exit gland **18**. In some embodiments, a strain relief member (e.g., a nylon bushing) may be provided at the cable exit gland **18** to reduce the amount of strain applied to the electrical wiring in response to rotational movement about the rotational axis.

(27) Unlike conventional lighting devices, the present invention provides both power and rotation through the same axis by positioning the rotational components of the hinge **13** (i.e., the hinge element **16** and end cap **17**) away from the rotational axis of the hinge mechanism. This is achieved, in one embodiment, by positioning the position holding gear of the hinge element **16** so that it travels around the semi-circular inner surface of the end cap **17** in an arc, whose radius is a fixed distance (d) away from the rotational axis of the hinge **13**.

(28) FIG. **4** is a photograph of various components that may be included within LED lamp **10**, such as a power supply board **20**, emitter housing **11**, emitter board **21**, 120×120 degree reflector **22**, 60×60 degree reflector **23**, and exit lens **24**. Although two reflectors are shown in the photograph of FIG. **4**, the assembled LED lamp **10** would include either the 120×120 degree reflector **22** or the 60×60 degree reflector **23**, but not both. Power supply board **20** connects the LED lamp **10** to the AC mains (not shown) and resides in power supply housing **12** (shown in FIG. **1**). Power supply board **20** provides DC power and control to emitter board **21**, which comprises the emitter modules and driver circuits. Emitter board **21** resides inside emitter housing **11** and is covered by either reflector **22** or reflector **23**. The exit lens **24** is mounted above the reflector **22**/**23** and attached to the sidewalls of the emitter housing **11**. As shown in FIG. **1**, the exit lens **24** is configured such that the external surface of the lens is substantially flush with the top of the sidewalls of the emitter housing. As described in more detail below, exit lens **24** may comprise an array of small lenses (or lenslets) on the external surface of the exit lens to improve color mixing and beam shape.

(29) FIGS. **1** and **4** illustrate one possible set of components for a linear LED lamp **10**, in accordance with the present invention. Other embodiments of linear LED lights could have substantially different components and/or dimensions for different applications. For instance, if LED lamp **10** was used for outdoor wall washing, the mechanics, optics and dimensions could be significantly different than those shown in FIGS. **1** and **4**. As such FIGS. **1** and **4** provide just one example of a linear LED lamp.

(30) FIG. **5** is an exemplary block diagram for the circuitry included on power supply board **20** and emitter board **21**. Power supply board **20** comprises AC/DC converter **30** and controller **31**. AC/DC converter **30** converters AC mains power to a DC voltage of typically 15-20V, which is then used to power controller **31** and emitter board **21**. Each such block may further regulate the DC voltage from AC/DC converter **30** to lower voltages as well. Controller **31** communicates with emitter board **21** through a digital control bus, in this example. Controller **31** could comprise a wireless,

powerline, or any other type of communication interface to enable the color of LED lamp **10** to be adjusted. In the illustrated embodiment, emitter board **21** comprises six emitter modules **33** and six interface circuits **32**. Interface circuits **32** communicate with controller **31** over the digital control bus and produce the drive currents supplied to the LEDs within the emitter modules **33**.

(31) FIG. **6** illustrates exemplary circuitry that may be included within interface circuitry **32** and emitter modules **33**. Interface circuitry **32** comprises control logic **34**, LED drivers **35**, and receiver **36**. Emitter module **33** comprises emission LEDs **37** and a detector **38**. Control logic **34** may comprise a microcontroller or special logic, and communicates with controller **31** over the digital control bus. Control logic **34** also sets the drive current produced by LED drivers **35** to adjust the color and/or intensity of the light produced by emission LEDs **37**, and manages receiver **36** to monitor the light produced by each individual LED **37** via detector **38**. In some embodiments, control logic **34** may comprise memory for storing calibration information necessary for maintaining precise color, or alternatively, such information could be stored in controller **31**.

(32) According to one embodiment, LED drivers **35** may comprise step down DC to DC converters that provide substantially constant current to the emission LEDs **37**. Emission LEDs **37**, in this example, may comprise white, blue, green, and red LEDs, but could include substantially any other combination of colors. LED drivers **35** typically supply different currents (levels or duty cycles) to each emission LED **37** to produce the desired overall color output from LED lamp **10**. In some embodiments, LED drivers **35** may measure the temperature of the emission LEDs **37** through mechanisms described, e.g., in pending U.S. patent application Ser. Nos. 13/970,944; 13/970,964; 13/970,990; and may periodically turn off all LEDs but one to perform optical measurements during a compensation period. The optical and temperature measurements obtained from the emission LEDs **37** may then be used to adjust the color and/or intensity of the light produced by the linear LED lamp **10** over time and with changes in temperature.

(33) FIG. **7** is an illustration of an exemplary color gamut produced with the red, green, blue, and white emission LEDs **37** included within linear LED lamp **10**. Points **40**, **41**, **42**, and **43** represent the color produced by the red, green, blue, and white LEDs **37** individually. The lines **44**, **45**, and **46** represent the boundaries of the colors that this example LED lamp **10** could produce. All colors within the triangle formed by **44**, **45**, and **46** can be produced by LED lamp **10**.

(34) FIG. **7** is just one example of a possible color gamut that can be produced with a particular combination of multi-colored LEDs. Alternative color gamuts can be produced with different LED color combinations. For instance, the green LED within LEDs **37** could be replaced with another phosphor converted LED to produce a higher lumen output over a smaller color gamut. Such phosphor converted LEDs could have a chromaticity in the range of (0.4, 0.5) which is commonly used in white plus red LED lamps. Additionally, cyan or yellow LEDs could be added to expand the color gamut. As such, FIG. **7** illustrates just one exemplary color gamut that could be produced with LED lamp **10**.

(35) Detector **38** may be any device, such as a silicon photodiode or an LED, that produces current indicative of incident light. In at least one embodiment, however, detector **38** is preferably an LED with a peak emission wavelength in the range of approximately 550 nm to 700 nm. A detector **38** with such a peak emission wavelength will not produce photocurrent in response to infrared light, which reduces interference from ambient light. In at least one preferred embodiment, detector **38** may comprise a small red, orange or yellow LED.

(36) Referring back to FIG. **6**, detector **38** is connected to a receiver **36**. Receiver **36** may comprise a trans-impedance amplifier that converts photocurrent to a voltage that may be digitized by an ADC and used by control logic **34** to adjust the drive currents, which are supplied to the emission LEDs **37** by the LED drivers **35**. In some embodiments, receiver **36** may further be used to measure the temperature of detector **38** through mechanisms described, e.g., in pending U.S. patent application Ser. Nos. 13/970,944, 13/970,964, 13/970,990. This temperature measurement may be used, in some embodiments, to adjust the color and/or intensity of the light produced by the linear

LED lamp **10** over changes in temperature.
(37) FIG. **5** and FIG. **6** are just examples of many possible block diagrams for power supply board **20**, emitter board **21**, interface circuitry **32**, and emitter module **33**. In other embodiments, interface circuitry **32** could be configured to drive more or less LEDs **37**, or may have multiple receiver channels. In yet other embodiments, emitter board **21** could be powered by a DC voltage, and as such, would not need AC/DC converter **30**. Emitter module **33** could have more or less LEDs **37** configured in more or less chains, or more or less LEDs per chain. As such, FIG. **5** and FIG. **6** are just examples.
(38) FIGS. **8-9** depict an exemplary emitter module **33** that may be used to improve color mixing in the linear LED lamp **10**. As shown in FIG. **8**, emitter module **33** may include an array of four emission LEDs **37** and a detector **38**, all of which are mounted on a common substrate **70** and encapsulated in a dome **71**. In one embodiment, the substrate **70** may be a ceramic substrate formed from an aluminum nitride or an aluminum oxide material (or some other reflective material) and may generally function to improve output efficiency by reflecting light back out of the emitter module **33**.
(39) The dome **71** may comprise substantially any optically transmissive material, such as silicone or the like, and may be formed through an overmolding process, for example. In some embodiments, a surface of the dome **71** may be lightly textured to increase light scattering and promote color mixing, as well as to reflect a small amount (e.g., about 5%) of the emitted light back toward the detector **38** mounted on the substrate **70**. The size of the dome **71** (i.e., the diameter of the dome in the plane of the LEDs) is generally dependent on the size of the LED array. However, it is generally desired that the diameter of the dome be substantially larger (e.g., about 1.5 to 4 times larger) than the diameter of the LED array to prevent occurrences of total internal reflection. As described in more detail below, the size and shape (or curvature) of the dome **71** is specifically designed to enhance color mixing between the plurality of emitter modules **33**.
(40) FIG. **9** depicts a side view of the emitter module **33** to illustrate a desired shape of the dome **71**, according to one embodiment of the invention. As noted above, conventional emitter modules typically include a dome with a hemispherical shape, in which the radius of the dome in the plane of the LED array is the same as the radius of the curvature of dome. As shown in FIG. **9**, dome **71** does not have the conventional hemispherical shape, and instead, is a much flatter or shallower dome. In general, the radius (r.sub.dome) of the shallow dome **71** in the plane of the LED array is approximately 20-30% larger than the radius (r.sub.curve) of the curvature of dome **71**.
(41) In one example, the radius (r.sub.dome) of the shallow dome **71** in the plane of the LEDs may be approximately 4.8 mm and the radius (r.sub.curve) of the dome curvature may be approximately 3.75 mm. The ratio of the two radii (4.8/3.75) is 1.28, which has been shown to provide the best balance between color mixing and efficiency for at least one particular combination and size of LEDs. However, one skilled in the art would understand how alternative radii and ratios may be used to achieve the same or similar color mixing results.
(42) By configuring the dome **71** with a substantially flatter shape, the dome **71** shown in FIGS. **8-9** allows a larger portion of the emitted light to emanate sideways from the emitter module **33**. Stated another way, a shallower dome **71** allows a significant portion of the emitted light to exit the dome at small angles (α.sub.side) relative to the horizontal plane of the LED array. In one example, the shallower dome **71** may allow approximately 40% of the light emitted by the array of LEDs **37** to exit the shallow dome at approximately 0 to 30 degrees relative to the horizontal plane of the LED array. In comparison, a conventional hemispherical dome may allow only 25% (or less) of the emitted light to exit between 0 and 30 degrees. As described in more detail below with reference to FIGS. **14-15**, the shallow dome **71** shown in FIGS. **8-9** improves color mixing in the linear LED lamp **10** by allowing a significant portion (e.g., 40%) of the light emitted from the sides of adjacent emitter modules to intermix before that light is reflected back out of the lamp.
(43) FIGS. **10**A-**10**B are exemplary drawings of the emitter module **33** shown in FIGS. **8-9**

including emission LEDs **37** and detector **38** within shallow dome **71**. As shown in FIGS. **10**A-**10**B, the four differently colored (e.g., red, green, blue and white) emission LEDs **37** are arranged in a square array and are placed as close as possible together in the center of the dome **71**, so as to approximate a centrally located point source. As noted above, it is generally desired that the diameter (d.sub.dome) of the dome **71** in the plane of the LEDs is substantially larger than the diameter (d.sub.array) of the LED array to prevent occurrences of total internal reflection. In one example, the diameter (d.sub.dome) of the dome **71** in the plane of the LEDs may be approximately 7.5 mm and the diameter (d.sub.array) of the LED array may be approximately 2.5 mm. Other dimensions may be appropriate in other embodiments of the invention.

(44) FIGS. **10**A-**10**B also illustrate exemplary placements of the detector **38** relative to the array of emission LEDs **37** within the shallow dome **71**. As shown in the embodiment of FIG. **10**A, the detector **38** may be placed closest to, and in the middle of, the edge of the array that is furthest from the short wavelength emitters. In this example, the short wavelength emitters are the green and blue LEDs positioned at the top of the array, and the detector **38** is an orange LED, which is least sensitive to blue light. Although somewhat counterintuitive, it is desirable to place the detector **38** as far away as possible from the blue LED so as to gather the most light reflected off the surface of the shallow dome **71** from the blue LED. As noted above, a surface of the dome **71** may be lightly textured, in some embodiments, so as to increase the amount of emitted light that is reflected back to the detector **38**.

(45) FIG. **10**B illustrates an alternative placement for the detector **38** within the shallow dome **71**. In some embodiments, the best place for the detector **38** to capture the most light from the blue LED may be on the other side of the array, and diagonally across from, the blue LED. In the embodiment shown in FIG. **10**B, the detector **38** is preferably placed somewhere between the dome **71** and a corner of the red LED. Since the green LED produces at least 10× the photocurrent as the blue LED on the orange detector, FIG. **10**B represents an ideal location for an orange detector **38** in relation to the particular RGBW array **37** described above. However, the detector **38** may be positioned as shown in FIG. **10**A, without sacrificing detection accuracy, if there is insufficient space between the dome **71** and the corner of the red LED, as shown in FIG. **10**B.

(46) FIG. **11** illustrates an exemplary emitter board **21** comprising six emitter modules **100**, **101**, **102**, **103**, **104**, and **105** arranged in a line. Each of the emitter modules shown in FIG. **11** may be identical to the emitter module **33** shown in FIGS. **8-10** and described above. FIG. **11** illustrates a preferred method for altering the orientation of emitter modules, or sets of emitter modules, to further improve color mixing there between. In the embodiment of FIG. **11**, the orientation of emitter modules **102** and **105** (i.e., a first set of emitter modules) is the same, the orientation of emitter modules **101** and **104** (i.e., a second set of emitter modules) is the same, and the orientation of emitter modules **100** and **103** (i.e., a third set of emitter modules) is the same. However, the orientation of the second set of emitter modules **101** and **104** is rotated 120 degrees from that of the first set of emitter modules **102** and **105**. Likewise, the orientation of the third set of emitter modules **100** and **103** is rotated 120 degrees from that of the second set of emitter modules **101** and **104**, and 240 degrees from the first set of emitter modules **102** and **105**. This rotation in combination with the shallow curvature of dome **71** enables the various colors of light produced by the plurality of emitter modules **100**, **101**, **102**, **103**, **104**, and **105** to thoroughly mix.

(47) FIG. **11** is just one example of an emitter board **21** that may be used to improve color mixing in a linear LED lamp **10**. Although the emitter board **21** is depicted in FIG. **11** with six emitter modules spaced approximately 2 inches apart, an emitter board **21** in accordance with the present invention could have substantially any number of emitter modules spaced substantially any distance apart. In embodiment shown in FIG. **11**, three sets of emitter modules are rotated 120 degrees from each other. In other embodiments, however, one or more of the emitter modules could be rotated by any amount provided that the emitter modules on the emitter board **21** make an integer number of rotations along the length of emitter board **21**.

(48) For example, each emitter module may be rotated an additional X degrees from a preceding emitter module in the line. Generally speaking, X is a rotational angle equal to 360 degrees divided by an integer N, where N is greater than or equal to 3. The number N is dependent on the number of emitter modules included on the emitter board. For instance, with six emitter modules, each module could be rotated 60 or 120 degrees from the preceding emitter module. With eight emitter modules, each module could be rotated an additional 45 or 90 degrees. For best color mixing, the rotational angle X should be equal to 360 degrees divided by three or four depending on how many emitter modules are included on the emitter board **21**.
(49) FIG. **12** is a photograph of the emitter board **21** and reflector **22** placed within the emitter housing **11** of the linear LED lamp **10**. In particular, FIG. **12** illustrates an exemplary placement of the emitter modules **33** and reflector **22** within emitter housing **11** for 120×120 degree beam applications. As noted above with regard to FIG. **11**, each set of emitter modules **33** (e.g., modules **102**/**105**, **101**/**104** and **100**/**103** shown in FIG. **11**) may be rotated 120 degrees relative to each other to improve color mixing. In the embodiment of FIG. **12**, the reflector **22** comprises a highly reflective material (e.g., vacuum metalized aluminum) that covers the entire inside of the emitter housing **11** except for the emitter modules **33**. The reflector **22** used in this embodiment improves the overall optical efficiency of the lamp **10** by reflecting light scattered off the exit lens The rotation of the emitter modules **33**, the shallow dome **71**, and the shape of the exit lens **24** (discussed below) all contribute to produce thorough color mixing throughout the 120×120 beam in this example.
(50) FIG. **13** is a photograph of the emitter board **21** and reflector **23** placed within the emitter housing **11**. In particular, FIG. **13** illustrates an exemplary placement of the emitter modules **33** and reflector **23** within emitter housing **11** for 60×60 degree beam applications. As in FIG. **12**, the sets of emitter modules **33** may be rotated 120 degrees relative to each other to improve color mixing. Like reflector **22**, reflector **23** also comprises a highly reflective material (e.g., vacuum metalized aluminum) to improve optical efficiency, however, reflector **23** additionally includes a plurality of louvers, each of which is centered around and suspended above a different one of the emitter modules **33**. As depicted more clearly in FIGS. **14-15**, the louvers are attached to the reflector **23** only on the sides and ends, and are open below. The space between the emitter modules **33** and the bottom of the louvers allows light emitted sideways from the emitter modules **33** to intermix to improve color uniformity in the output beam.
(51) FIG. **14** is an exemplary ray diagram illustrating the color mixing effect between emitter modules **100-105** and reflector **23**. As shown in FIG. **14**, louvers **110**, **111**, **112**, **113**, **114**, and **115** are individually centered upon and positioned above a different emitter module. The louvers **110-115** focus a majority of the light emitted from the emitter modules **100-105** into an output beam, but allow some of the light that emanates from the side of the emitter modules **100-105** to mix with light from other emitter modules. For example, louver **112** focuses most of the light emitted from emitter module **102** into the output beam, however, some rays from emitter module **102** are reflected by louvers **111**, **113**, and **115**. Likewise, louver **113** focuses most of the light emitted from emitter module **103**; however, some rays from emitter module **103** are reflected by louvers **110**, **112**, and **114**. The exemplary ray diagram of FIG. **14** illustrates only a limited number of rays. In reality, each louver **110-115** reflects some light from all emitter modules **100-105**, which significantly improves color mixing in the resulting beam.
(52) FIG. **15** illustrates a cross section of a portion of the exemplary 60×60 degree reflector **23** comprising louver **110** and emitter module **100**. Louver **110** is attached to both lateral sides of reflector **23**. The same is true for louvers **111-115**. Additionally, louvers **110** and **115** are attached to the ends of reflector **23**. In some embodiments, the louvers **110-115** may be attached to the sidewalls and ends of the reflector **23** by forming the louvers and reflector as one integral piece (e.g., by a molding process). Other means for attachment may be used in other embodiments of the invention.

(53) The overall shape and size of the louvers **110-115** determine the shape, and to some extent the color, of the output beam. As shown in FIGS. **13-15**, each louver has a substantially round or circular shape with sloping sidewalls. As shown in FIG. **15**, the sidewalls of the louvers are angled outward, such that the diameter at the bottom of the louver (d.sub.bottom) is substantially smaller than the diameter at the top of the louver (d.sub.top). It is generally desired that the louvers **110-115** be substantially larger than the emitter modules **100-105**, so that the louvers may focus a majority of the light emitted by the emitter modules into an output beam. As noted above, the diameter of the emitter module (d.sub.emit) may be about 7.5 mm, in one embodiment. In such an embodiment, the bottom diameter (d.sub.bottom) of the louver may be about 35 mm and the top diameter (d.sub.top) of the louver may be about 42 mm. Other dimensions and shapes may be appropriate in other embodiments of the invention. In one alternative embodiment, for example, the louvers may alternatively be configured with a substantially parabolic shape, as would be appropriate in 30×60 beam applications.
(54) As further depicted in FIG. **15**, the angle (α.sub.ref) of the sidewalls of reflector **23** is substantially the same as the angle (α.sub.ref) of the sidewalls of the louvers **110-115**. According to one embodiment, the angle of the sidewall surfaces of the reflector **23** and the angle of the louvers **110-115** may be approximately 60 degrees. In the illustrated embodiment, the shape and size of the reflector and louvers are chosen for 60×60 beam applications. One skilled in the art would understand how alternative shapes and sizes may be used to produce other beam shapes. As such, FIGS. **13-15** are just example illustrations of the invention.
(55) As further shown in FIG. **15**, the louvers (e.g., **110**) are formed so as to include a plurality of planar facets, or lunes **116**, in the sidewalls. Lunes **116** are flattened segments in the otherwise round louvers **110-115**. The lunes **116** generally function to randomize the direction of the light rays and improve color mixing. FIG. **15** further depicts how the louvers (e.g., **110**) are suspended some height (h) above the emitter modules (e.g., **100**). The height (h) is generally dependent on the shape of the shallow dome **71** and the configuration of the lunes **116**. According to one embodiment, the louvers **110-115** may be suspended approximately 5 mm to approximately 10 mm above the emitter modules **100-105** to allow a sufficient amount of light to mix underneath the louvers.
(56) In addition the features described above (e.g., the flattened dome shape, the rotated emitter modules, the reflector with floating louvers, etc.), the exit lens **24** of the linear LED lamp **10** provides an additional measure of color mixing and beam shaping for the output beam. In general, the exit lens **24** is preferably configured with some combination of differently textured surfaces and/or patterns on opposing sides of the exit lens. The exit lens **24** preferably comprises injection modeled PMMA (acrylic), but could comprise substantially any other optically transparent material.
(57) FIGS. **16** and **17** illustrate one exemplary embodiment of an exit lens **24** comprising an internal surface having a flat roughened surface that diffuses the light passing through the exit lens, and an array of micro-lenses or lenslets **120** formed on an external surface of the lens. As shown in FIG. **16**, the lenslets **120** may be rectangular or square-shaped domes, and may be approximately 1 mm square, but could have a variety of other shapes and sizes. The curvature of lenslets **120** is defined by the radius of the arcs that create the lenslets. In one embodiment, the radius of the lenslets **120** is about 1 mm. Although any combination of size, shape and curvature of lenslets **120** is possible, such dimensions have been shown to provide optimum color mixing and beam shaping performance.
(58) FIG. **16** is just one example of an exit lens **24**. One skilled in the art would understand how an exit lens may be alternatively configured to produce the same or similar color mixing results. In other embodiments, for example, the pattern on the exterior surface of the exit lens could be hexagonal instead of rectangular, and/or the diameter of the lenslets **120** could be different. Likewise, the curvature of the lenslets **120** could change significantly and still achieve the desired

results. In general, the exit lens **24** described herein may provide improved color mixing with substantially any shape, any diameter, and any lenslet curvature by providing an array of lenslets on at least one side of the exit lens **24**. In some embodiments, an array of similarly or differently configured lenslets may also be provided on the interior surface of the exit lens.
(59) FIG. **17** illustrates a ray diagram for the exemplary exit lens **24** shown in FIG. **16**. In this example, the light rays **130** from the emitter modules **33** enter the exit lens **24** through the flat roughened internal side and are diffused within the exit lens **24**. The scattered light rays within the exit lens **24** are further randomized by the array of lenslets **120** formed on the external side of the exit lens to produce an output beam **131** with substantially uniform color throughout the beam.
(60) It will be appreciated to those skilled in the art having the benefit of this disclosure that this invention is believed to provide color mixing optics and optical feedback to produce uniform color throughout the output light beam of a multi-color linear LED illumination device. More specifically, the invention provides an emitter module comprising a plurality of emission LEDs and a detector LED, all of which are mounted on a substrate and encapsulated in a shallow dome. The shallow dome allows a significant portion of the emitted light to emanate from the side of the emitter module, where it can mix with light from other emitter modules to improve color mixing. The invention further improves color mixing within a multi-color linear LED illumination device by rotating sets of the emitter modules relative to each other and providing a reflector comprising a plurality of floating louvers, which are centered upon and suspended above each of the emitter modules. The floating louvers allow a portion of the light emitted from each emitter module to mix with light from other emitter modules to produce uniform color throughout the resulting output beam. Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. It is intended that the following claims be interpreted to embrace all such modifications and changes and, accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

Claims

.[.1. An illumination device, comprising: a plurality of emitter modules spaced apart from each other and arranged in a line, wherein each emitter module comprises an array of at least two different colors of light emitting diodes (LEDs), which are mounted on a substrate and encapsulated within a shallow dome, and wherein a flattened shape of the shallow dome allows a greater portion of light emitted by the array of LEDs to emanate sideways from the emitter module than a hemispherical shaped dome; and a reflector comprising a plurality of louvers, wherein each louver is centered upon and suspended a spaced distance above a different one of the emitter modules to focus a majority of light emitted by that emitter module into an output beam, and wherein each louver is configured to reflect the portion of the light that emanates sideways from adjacent emitter modules to improve color mixing in the output beam..].
.[.2. The illumination device as recited in claim 1, wherein a radius of the shallow dome in a plane of the array of LEDs is 20-30% larger than a radius of a curvature of the shallow dome, so that the portion of the light that emanates sideways from the emitter module exits the shallow dome at small angles relative to a plane of the LED array..].
.[.3. The illumination device as recited in claim 2, wherein approximately 40% of the light emitted by the array of LEDs exits the shallow dome at approximately 0 to 30 degrees relative to the plane of the LED array..].
.[.4. The illumination device as recited in claim 1, wherein a top diameter of each louver is substantially larger than a bottom diameter of the louver..].
.[.5. The illumination device as recited in claim 4, wherein the plurality of louvers each comprise a substantially circular shape with sloping sidewalls..].
.[.6. The illumination device as recited in claim 4, wherein the plurality of louvers each comprise

sidewalls with a substantially parabolic shape..].
.[.7. The illumination device as recited in claim 4, wherein the louvers are configured to focus the majority of the light emitted by the emitter modules into the output beam by configuring the bottom diameter of the louvers to be substantially larger than a diameter of the emitter modules..].
.[.8. The illumination device as recited in claim 4, wherein the sloping sidewalls of the louvers include a plurality of planar facets, which are configured to randomize a direction of light reflected from the planar facets..].
.[.9. The illumination device as recited in claim 4, wherein the louvers are suspended approximately 5 mm to approximately 10 mm above the emitter modules to allow the portion of the light that emanates sideways from the emitter modules to mix underneath the louvers..].
.[.10. The illumination device as recited in claim 1, wherein the plurality of emitter modules comprise at least a first emitter module, a second emitter module, and a third emitter module, and wherein: the second emitter module is spaced equally distant between the first and third emitter modules; the second emitter module is rotated X degrees relative to the first emitter module; the third emitter module is rotated 2X degrees relative to the first emitter module; and wherein X is a rotational angle equal to 360 degrees divided by an integer N, where N is greater than or equal to 3..].
.[.11. The illumination device as recited in claim 1, wherein the array of LEDs comprises at least four LEDs, which are mounted on the substrate close together and arranged in a square pattern near a center of the shallow dome..].
.[.12. The illumination device as recited in claim 11, wherein the array of LEDs comprises a red LED, a green LED, a blue LED and a white LED..].
.[.13. The illumination device as recited in claim 1, further comprising: an emitter housing, wherein the plurality of emitter modules and the reflector reside within the emitter housing; and an exit lens mounted above the reflector and attached to sidewalls of the emitter housing..].
.[.14. The illumination device as recited in claim 13, wherein an internal surface of the exit lens comprises a flat roughened surface that scatters light rays passing through the exit lens, and wherein an external surface of the exit lens comprises an array of lenslets that randomizes the scattered light rays..].
.Iadd.15. A light-emitting diode (LED) illumination device, comprising: a mounting bracket for attaching the illumination device to a surface; a swing arm extending from the mounting bracket, a first end of the swing arm engaging the mounting bracket and an opposite second end of the swing arm having an opening for receiving a power cable to power the illumination device; an end cap rotatably coupled to the second end of the swing arm adjacent to the opening of the swing arm, the end cap and the swing arm cooperating to define a rotational axis, wherein the rotational axis is shared by an axis defined by a route of the power cable, and wherein the end cap has a plurality of teeth disposed a radial distance from the rotational axis and arranged in a semi-circle; and a hinge element extending from an intermediate location on the swing arm in a position between the first and second ends, a portion of the hinge element engaging the teeth to provide a resistance to secure the illumination device in a rotational position with respect to the mounting bracket around the rotational axis..Iaddend.
.Iadd.16. The illumination device of claim 15, wherein the hinge element provides sufficient resistance to secure the illumination device in substantially any rotational position within a 180 degree range of motion..Iaddend.
.Iadd.17. The illumination device of claim 15, wherein the hinge element and the end cap form a rack and pinion gear assembly..Iaddend.
.Iadd.18. The illumination device of claim 15, wherein the hinge element is a position holding gear..Iaddend.
.Iadd.19. The illumination device of claim 18, wherein the hinge element further comprises a torque element that provides a substantially consistent amount of torque to the position holding

gear..Iaddend.
.Iadd.20. The illumination device of claim 19, wherein the torque element provides the substantially consistent amount of torque to the position holding gear when the position holding gear is stationary..Iaddend.
.Iadd.21. The illumination device of claim 19, wherein the torque element provides the substantially consistent amount of torque when the hinge element is rotated..Iaddend.
.Iadd.22. The illumination device of claim 15, wherein the hinge element comprises a pivotable, toothed rotatable gear, wherein the teeth on the hinge element engage the teeth on the end cap..Iaddend.
.Iadd.23. The illumination device of claim 15, wherein the hinge element further comprises a torque element that facilitates rotational adjustment the illumination device, while still providing the necessary resistance to hold the illumination device in a desired rotational position..Iaddend.
.Iadd.24. The illumination device of claim 23, wherein the torque element exhibits a high static energy and a low kinetic energy..Iaddend.
.Iadd.25. The illumination device of claim 15, further comprising a strain relief member disposed in the opening to reduce strain applied to the power cable in response to rotational movement about the rotational axis..Iaddend.
.Iadd.26. A light-emitting diode (LED) illumination device, comprising: a mounting bracket for attaching the illumination device to a surface; a swing arm extending from the mounting bracket, a first end of the swing arm engaging the mounting bracket and an opposite second end of the swing arm having an opening for receiving a power cable to power the illumination device; an end cap rotatably coupled to the second end of the swing arm adjacent to the opening of the swing arm, the end cap and the swing arm cooperating to define a rotational axis, wherein the rotational axis is shared by an axis defined by a route of the power cable, and wherein the end cap has a plurality of teeth arranged radially at distance from the rotational axis; a pivotable, toothed, hinge element extending from an intermediate location on the swing arm between the first and second ends, wherein the teeth on the hinge element engage the teeth on the end cap to provide a resistance to secure the illumination device in a rotational position with respect to the mounting bracket around the rotational axis; a housing physically coupled to the end cap; and a printed circuit board disposed at least partially within the housing, the printed circuit board including a plurality of LEDs..Iaddend.
.Iadd.27. The illumination device of claim 26, wherein the plurality of LEDs comprise a plurality of LED arrays disposed on a front surface of the printed circuit board..Iaddend.
.Iadd.28. The illumination device of claim 27, wherein each of the plurality of LED arrays comprise a 2×2 LED array..Iaddend.
.Iadd.29. The illumination device of claim 28, wherein the 2×2 LED array comprises four LEDs, each having a different output spectrum..Iaddend.
.Iadd.30. The illumination device of claim 29, wherein each of the plurality of 2×2 LED arrays is rotated by an angle with respect to a neighboring 2×2 LED array..Iaddend.
.Iadd.31. The illumination device of claim 28, wherein each of the 2×2 LED arrays comprise a red LED, a green LED, a blue LED, and a phosphor coated blue LED..Iaddend.
.Iadd.32. The illumination device of claim 31, further comprising a dome-shaped optical element disposed on each respective one of the plurality of LED arrays..Iaddend.
.Iadd.33. The illumination device of claim 32, wherein each of the LED arrays include a floating louver assembly disposed a distance from the dome-shaped optical element..Iaddend.
.Iadd.34. The illumination device of claim 26, wherein the hinge element provides sufficient resistance to secure the illumination device in substantially any rotational position within a 180 degree range of motion..Iaddend.
.Iadd.35. illumination device of claim 26, wherein the hinge element further comprises a position

holding gear and a torque element that provides a substantially consistent amount of torque to the position holding gear..Iaddend.